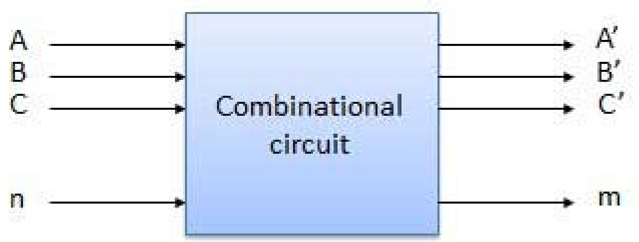
Combinational Logic Circuit

Combinational circuit is a circuit in which we combine the different gates in the circuit, for example encoder, decoder, multiplexer and demultiplexer. Some of the characteristics of combinational circuits are following −

- The output of combinational circuit at any instant of time, depends only on the levels present at input terminals.
- The combinational circuit do not use any memory. The previous state of input does not have any effect on the present state of the circuit.
- A combinational circuit can have an n number of inputs and m number of outputs.

Block diagram

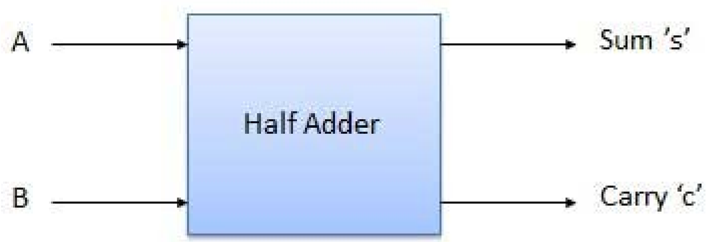


We're going to elaborate few important combinational circuits as follows.

Half Adder

Half adder is a combinational logic circuit with two inputs and two outputs. The half adder circuit is designed to add two single bit binary number A and B. It is the basic building block for addition of two **single** bit numbers. This circuit has two outputs **carry** and **sum**.

Block diagram



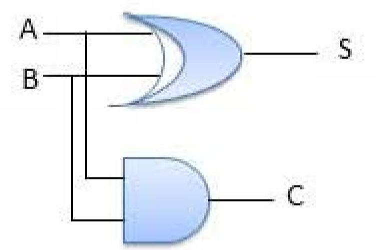
Truth Table

Inputs		Output	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

S=A’B+AB’

C=A.B

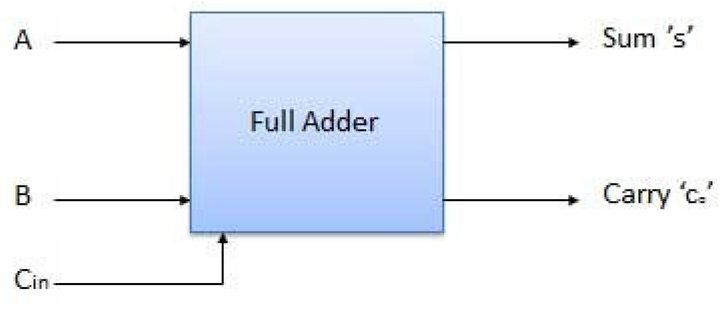
Circuit Diagram



Full Adder

Full adder is developed to overcome the drawback of Half Adder circuit. It can add two one-bit numbers A and B, and carry c. The full adder is a three input and two output combinational circuit.

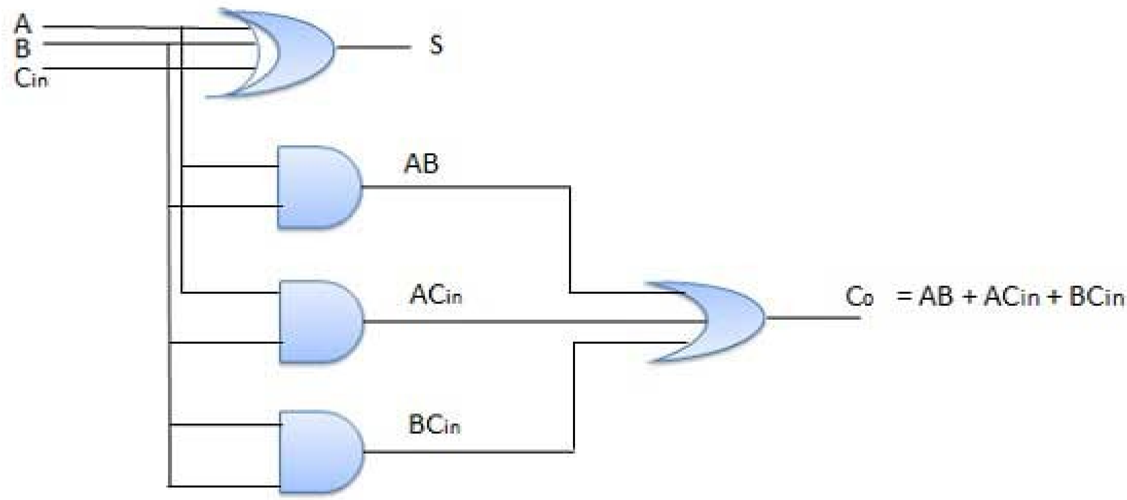
Block diagram



Truth Table

Inputs			Output	
A	B	C_{in}	S	C_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Circuit Diagram



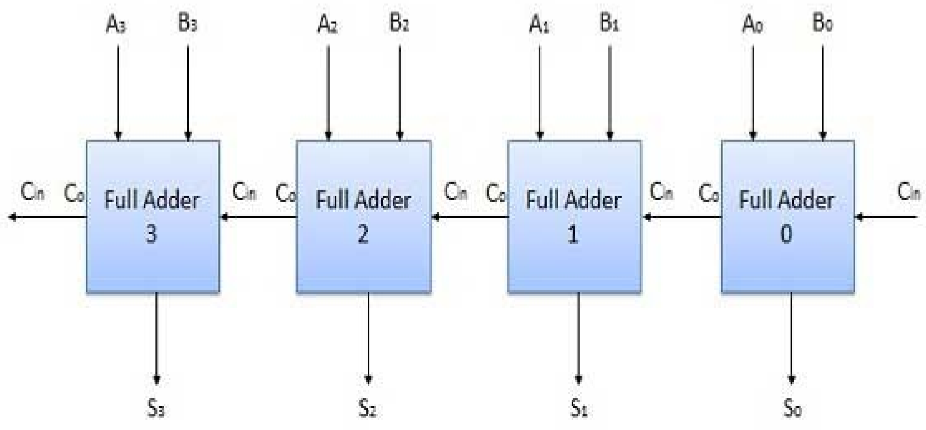
N-Bit Parallel Adder

The Full Adder is capable of adding only two single digit binary number along with a carry input. But in practical we need to add binary numbers which are much longer than just one bit. To add two n-bit binary numbers we need to use the n-bit parallel adder. It uses a number of full adders in cascade. The carry output of the previous full adder is connected to carry input of the next full adder.

4 Bit Parallel Adder

In the block diagram, A_0 and B_0 represent the LSB of the four bit words A and B. Hence Full Adder-0 is the lowest stage. Hence its C_{in} has been permanently made 0. The rest of the connections are exactly same as those of n-bit parallel adder is shown in fig. The four bit parallel adder is a very common logic circuit.

Block diagram



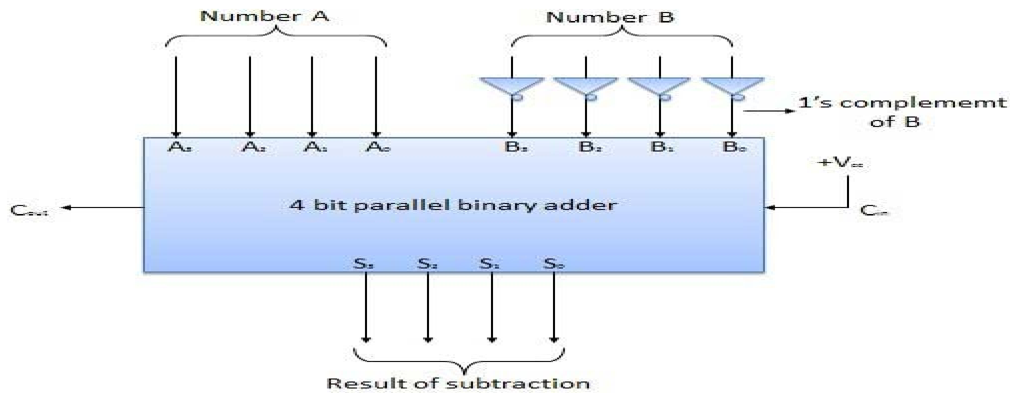
N-Bit Parallel Subtractor

The subtraction can be carried out by taking the 1's or 2's complement of the number to be subtracted. For example we can perform the subtraction (A-B) by adding either 1's or 2's complement of B to A. That means we can use a binary adder to perform the binary subtraction.

4 Bit Parallel Subtractor

The number to be subtracted (B) is first passed through inverters to obtain its 1's complement. The 4-bit adder then adds A and 2's complement of B to produce the subtraction. S_3 S_2 S_1 S_0 represents the result of binary subtraction (A-B) and carry output C_{out} represents the polarity of the result. If A > B then Cout = 0 and the result of binary form (A-B) then C_{out} = 1 and the result is in the 2's complement form.

Block diagram



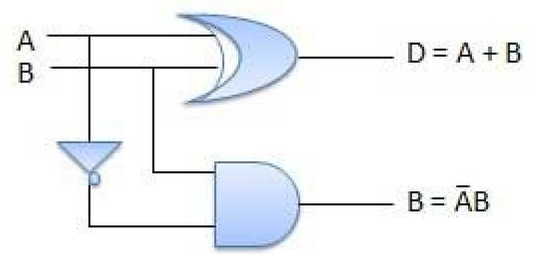
Half Subtractors

Half subtractor is a combination circuit with two inputs and two outputs (difference and borrow). It produces the difference between the two binary bits at the input and also produces an output (Borrow) to indicate if a 1 has been borrowed. In the subtraction (A-B), A is called as Minuend bit and B is called as Subtrahend bit.

Truth Table

Inputs		Output	
A	B	(A – B)	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Circuit Diagram



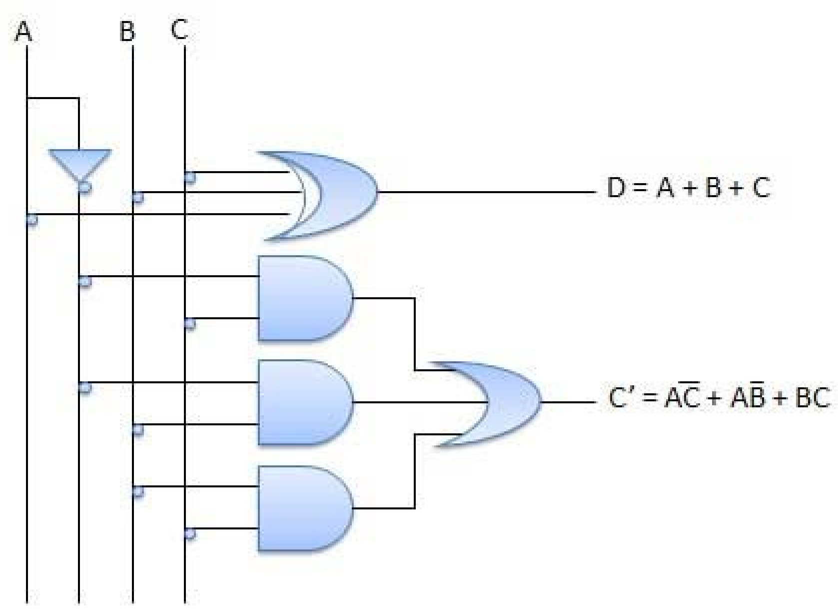
Full Subtractors

The disadvantage of a half subtractor is overcome by full subtractor. The full subtractor is a combinational circuit with three inputs A,B,C and two output D and C'. A is the 'minuend', B is 'subtrahend', C is the 'borrow' produced by the previous stage, D is the difference output and C' is the borrow output.

Truth Table

Inputs			Output	
A	B	C	(A-B-C)	C'
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

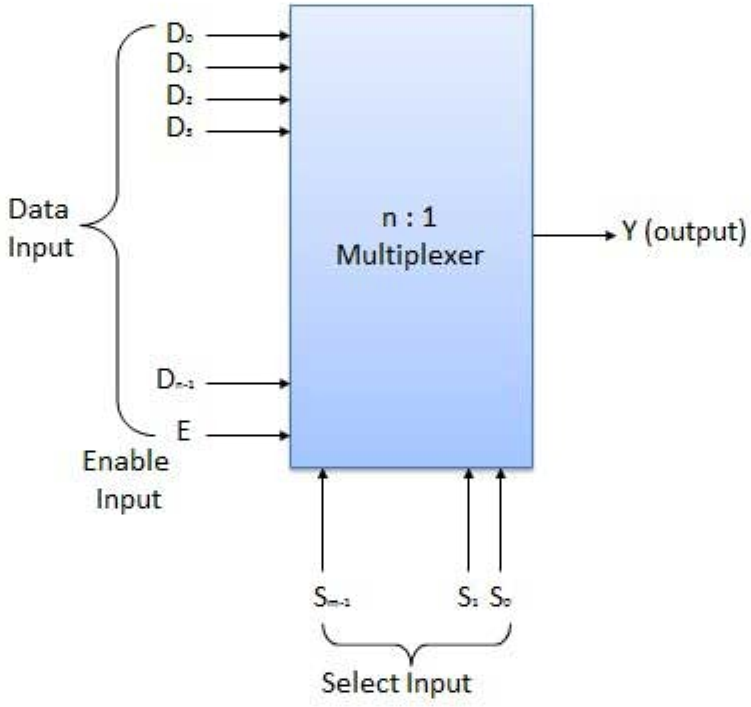
Circuit Diagram



Multiplexers

Multiplexer is a special type of combinational circuit. There are n-data inputs, one output and m select inputs with 2m = n. It is a digital circuit which selects one of the n data inputs and routes it to the output. The selection of one of the n inputs is done by the selected inputs. Depending on the digital code applied at the selected inputs, one out of n data sources is selected and transmitted to the single output Y. E is called the strobe or enable input which is useful for the cascading. It is generally an active low terminal that means it will perform the required operation when it is low.

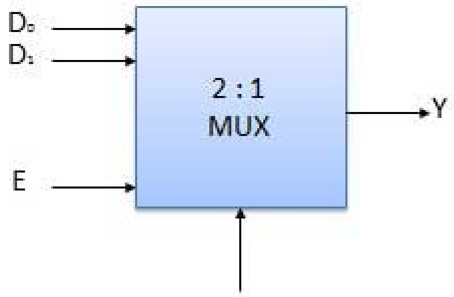
Block diagram



Multiplexers come in multiple variations

- 2 : 1 multiplexer
- 4 : 1 multiplexer
- 16 : 1 multiplexer
- 32 : 1 multiplexer

Block Diagram



Truth Table

Enable	Select	Output
E	S	Y
0	x	0
1	0	D_0
1	1	D_1

x = Don't care

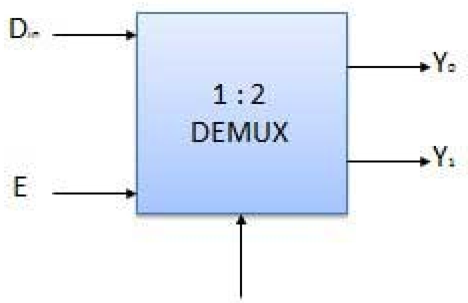
Demultiplexers

A demultiplexer performs the reverse operation of a multiplexer i.e. it receives one input and distributes it over several outputs. It has only one input, n outputs, m select input. At a time only one output line is selected by the select lines and the input is transmitted to the selected output line. A de-multiplexer is equivalent to a single pole multiple way switch as shown in fig.

Demultiplexers comes in multiple variations.

- 1 : 2 demultiplexer
- 1 : 4 demultiplexer
- 1 : 16 demultiplexer
- 1 : 32 demultiplexer

Block diagram



Truth Table

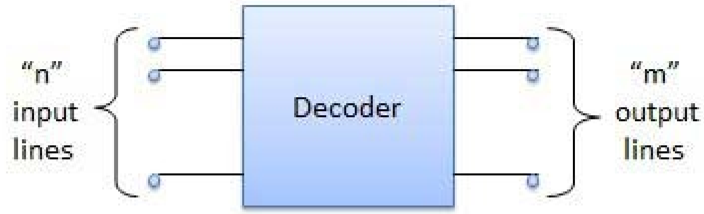
Enable	Select	Output	
E	S	Y0	Y1
0	x	0	0
1	0	0	D_{in}
1	1	D_{in}	0

x = Don't care

Decoder

A decoder is a combinational circuit. It has n input and to a maximum m = 2n outputs. Decoder is identical to a demultiplexer without any data input. It performs operations which are exactly opposite to those of an encoder.

Block diagram



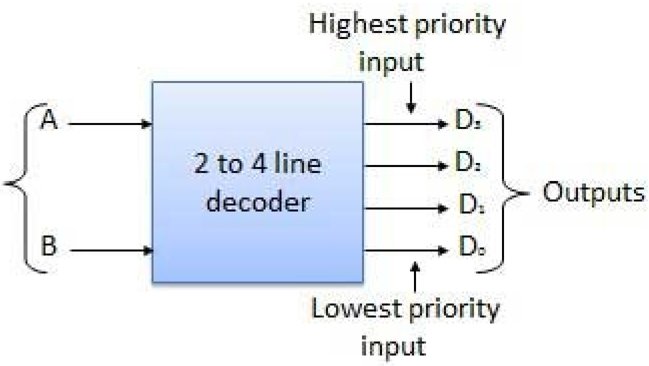
Examples of Decoders are following.

- Code converters
- BCD to seven segment decoders
- Nixie tube decoders
- Relay actuator

2 to 4 Line Decoder

The block diagram of 2 to 4 line decoder is shown in the fig. A and B are the two inputs where D through D are the four outputs. Truth table explains the operations of a decoder. It shows that each output is 1 for only a specific combination of inputs.

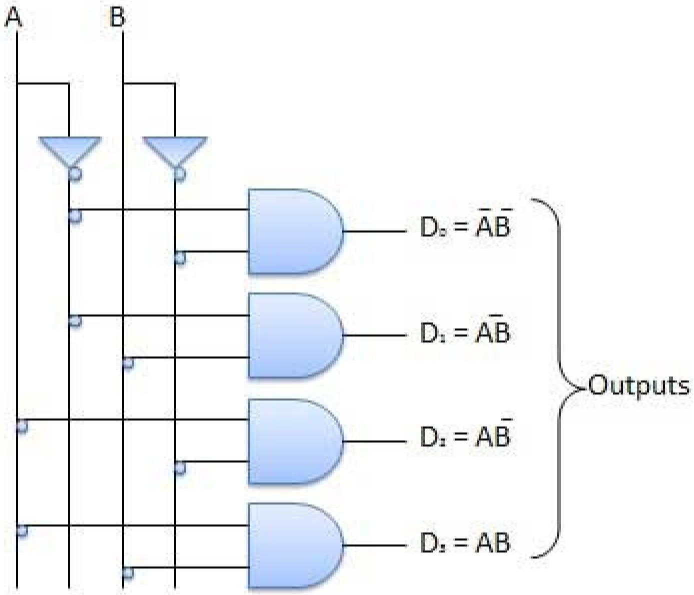
Block diagram



Truth Table

Inputs		Output			
A	B	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
0	1	0	0	1	0
1	1	0	0	0	1

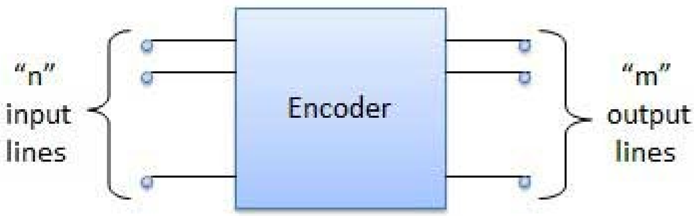
Logic Circuit



Encoder

Encoder is a combinational circuit which is designed to perform the inverse operation of the decoder. An encoder has n number of input lines and m number of output lines. An encoder produces an m bit binary code corresponding to the digital input number. The encoder accepts an n input digital word and converts it into an m bit another digital word.

Block diagram



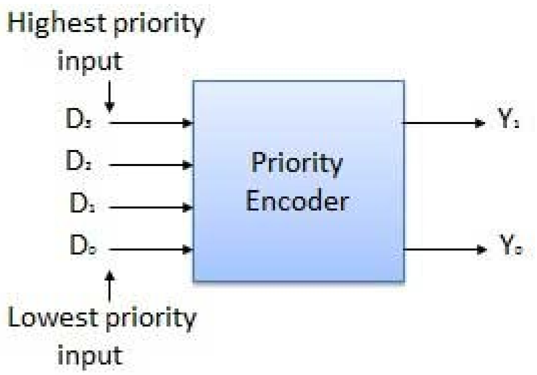
Examples of Encoders are following.

- Priority encoders
- Decimal to BCD encoder
- Octal to binary encoder
- Hexadecimal to binary encoder

Priority Encoder

This is a special type of encoder. Priority is given to the input lines. If two or more input line are 1 at the same time, then the input line with highest priority will be considered. There are four input D_0, D_1, D_2, D_3 and two output Y_0, Y_1. Out of the four input D_3 has the highest priority and D_0 has the lowest priority. That means if $D_3 = 1$ then Y_1 $Y_1 = 11$ irrespective of the other inputs. Similarly if $D_3 = 0$ and $D_2 = 1$ then Y_1 $Y_0 = 10$ irrespective of the other inputs.

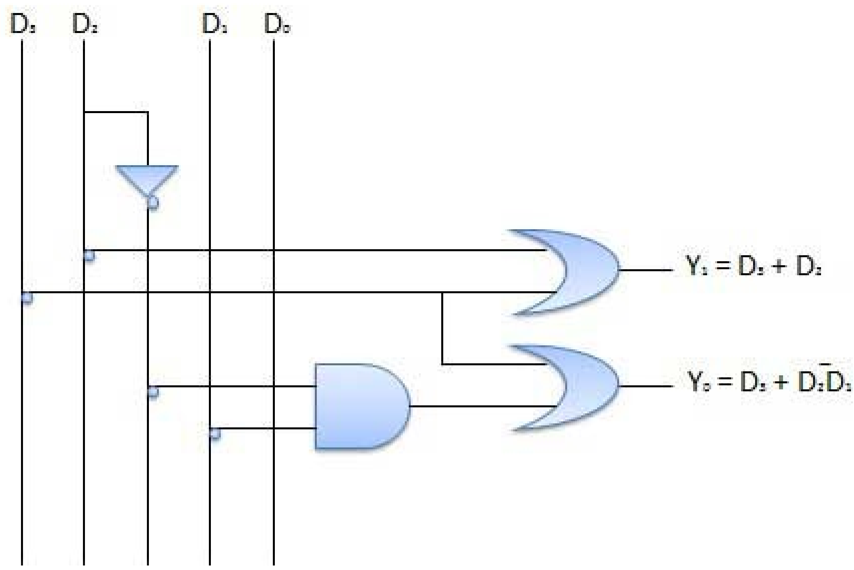
Block diagram



Truth Table

Highest	Inputs		Lowest	Outputs	
D_3	D_2	D_1	D_0	Y_0	Y_1
0	0	0	0	x	x
0	0	0	1	0	0
0	0	1	x	0	1
0	1	x	x	1	0
1	x	x	x	1	1

Logic Circuit



Half Adder-

- Half Adder is a combinational logic circuit.
- It is used for the purpose of adding two single bit numbers.
- It contains 2 inputs and 2 outputs (sum and carry).

Half Adder Designing-

Half adder is designed in the following steps-

Step-01:

Identify the input and output variables-

- Input variables = A, B (either 0 or 1)
- Output variables = S, C where S = Sum and C = Carry

Step-02:

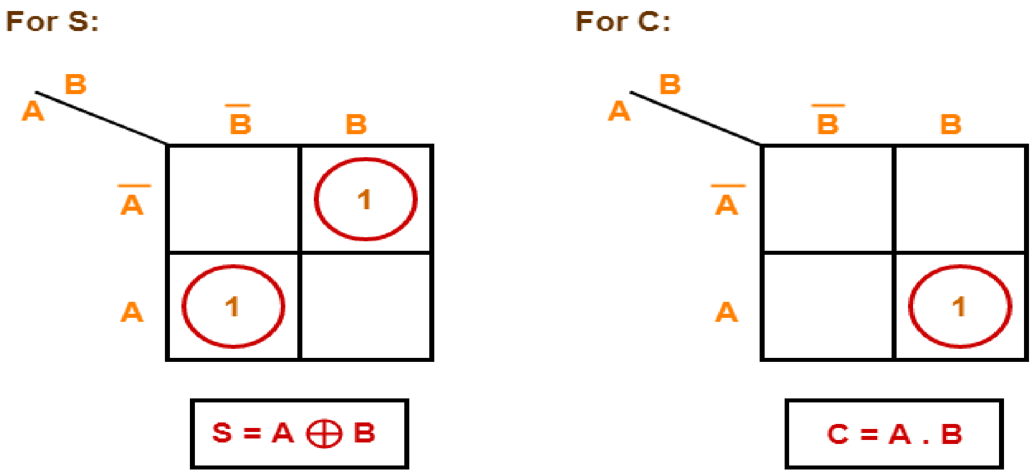
Draw the truth table-

Inputs		Outputs	
A	**B**	**S (Sum)**	**C (Carry)**
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Truth Table

Step-03:

Draw K-maps using the above truth table and determine the simplified Boolean expressions-

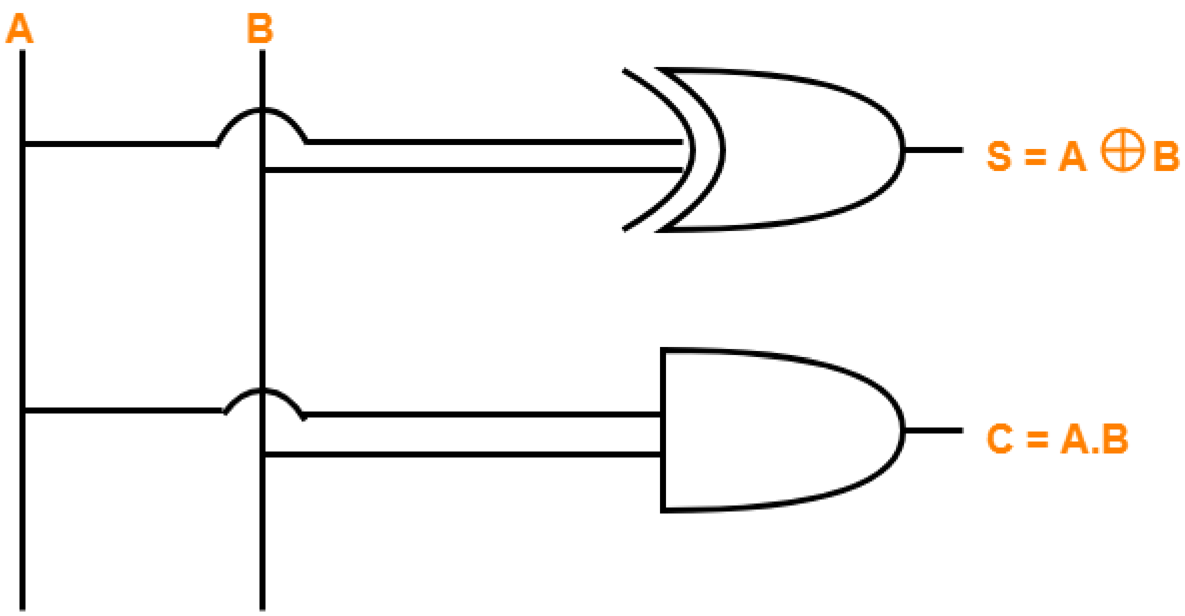


K Maps

Step-04:

Draw the logic diagram.

The implementation of half adder using 1 XOR gate and 1 AND gate is as shown below-



Half Adder Logic Diagram

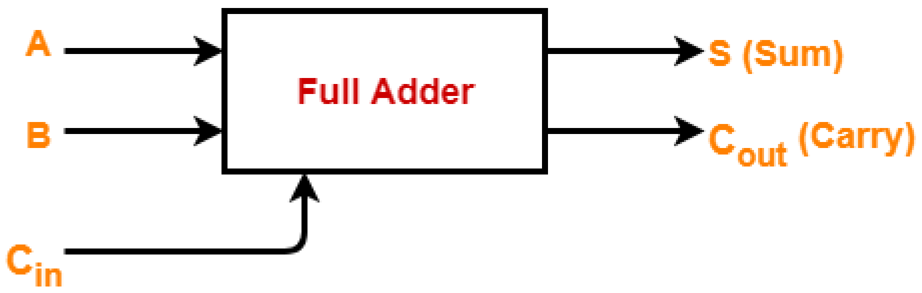
Limitation of Half Adder-

- Half adders have no scope of adding the carry bit resulting from the addition of previous bits.
- This is a major drawback of half adders.
- This is because real time scenarios involve adding the multiple number of bits which can not be accomplished using half adders.

To overcome this drawback, Full Adder comes into play.

Full Adder-

- Full Adder is a combinational logic circuit.
- It is used for the purpose of adding two single bit numbers with a carry.
- Thus, full adder has the ability to perform the addition of three bits.
- Full adder contains 3 inputs and 2 outputs (sum and carry) as shown-



Full Adder Designing-

Full adder is designed in the following steps-

Step-01:

Identify the input and output variables-

- Input variables = A, B, C_{in} (either 0 or 1)
- Output variables = S, C_{out} where S = Sum and C_{out} = Carry

Step-02:

Draw the truth table-

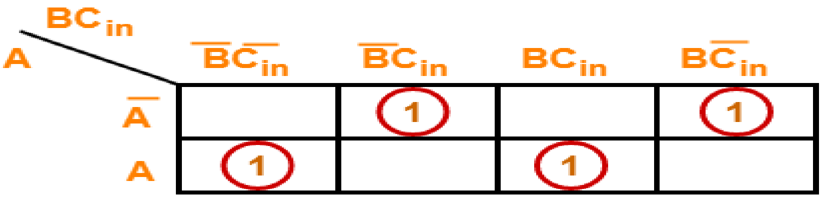
Inputs			**Outputs**	
A	**B**	**C_{in}**	**S (Sum)**	**C_{out} (Carry)**
0	0	0	0	0
0	0	1	1A’B’C_{in}	0
0	1	0	1 A’BC’	0
0	1	1	0	1A’BC
1	0	0	1AB’C’	0
1	0	1	0	1AB’C
1	1	0	0	1ABC’
1	1	1	1ABC	1ABC

Truth Table **S=A**

Step-03:

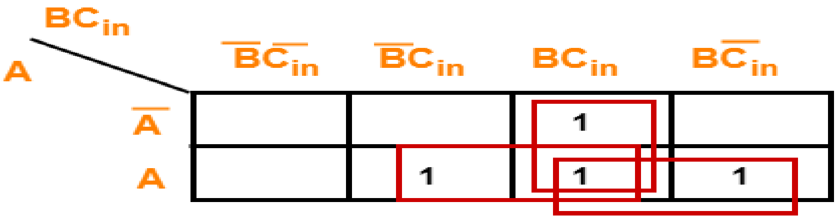
Draw K-maps using the above truth table and determine the simplified Boolean expressions-

For S:



$$\boxed{S = A \oplus B \oplus C_{in}}$$

For C_{in}:

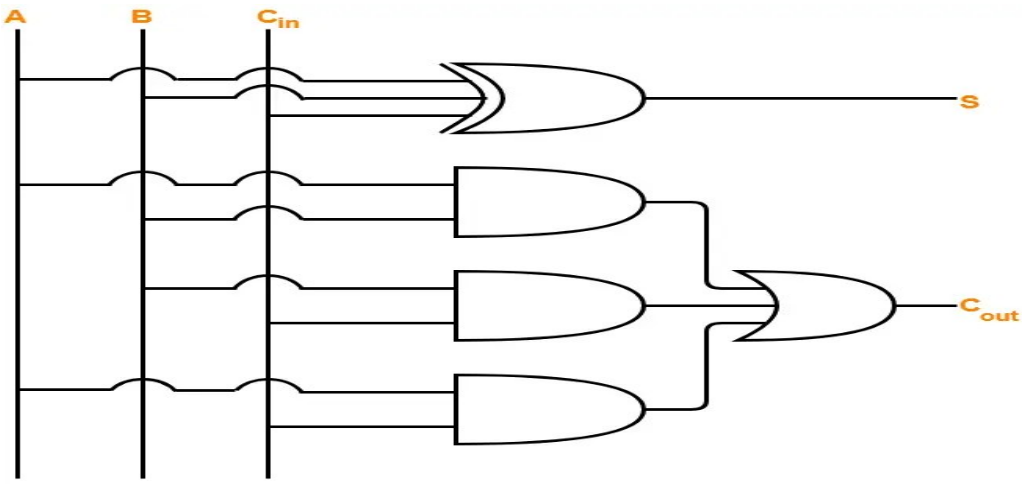


$$\boxed{C_{out} = AB + BC_{in} + C_{in}A}$$

Step-04:

Draw the logic diagram.

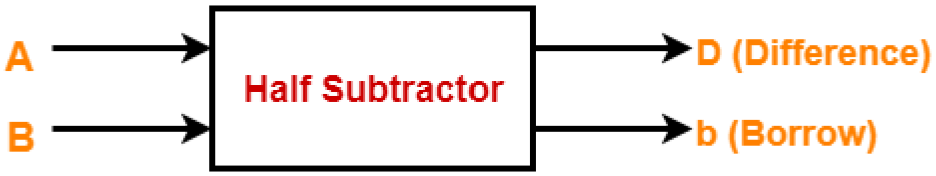
The implementation of full adder using 1 XOR gate, 3 AND gates and 1 OR gate is as shown below-



Full Adder Logic Diagram

Half Subtractor-

- Half Subtractor is a combinational logic circuit.
- It is used for the purpose of subtracting two single bit numbers.
- It contains 2 inputs and 2 outputs (difference and borrow).



Half Subtractor Designing-

Half subtractor is designed in the following steps-

Step-01:

Identify the input and output variables-

- Input variables = A, B (either 0 or 1)
- Output variables = D, b where D = Difference and b = borrow

Step-02:

Draw the truth table-

Inputs		Outputs	
A	**B**	**D (Difference)**	**b (Borrow)**
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Truth Table

D=A’B+AB’ B=A’B

0-0=0

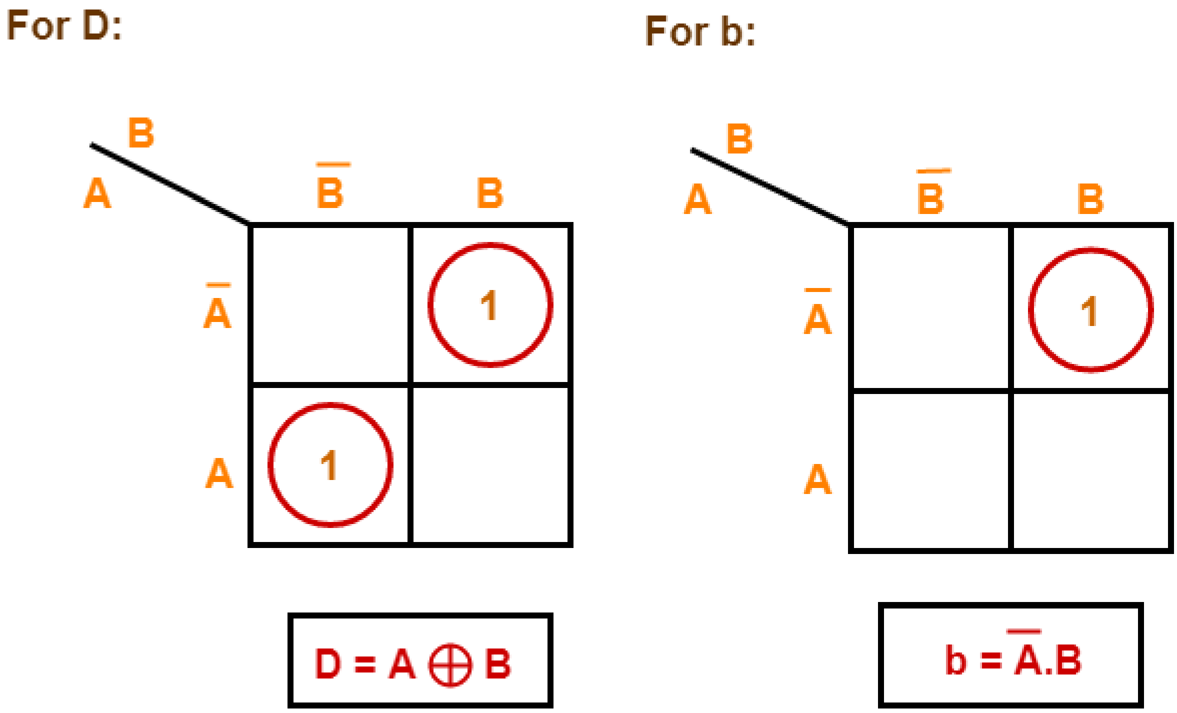
1-0=1

1-1=0

0-1=1d 1b

Step-03:

Draw K-maps using the above truth table and determine the simplified Boolean expressions-

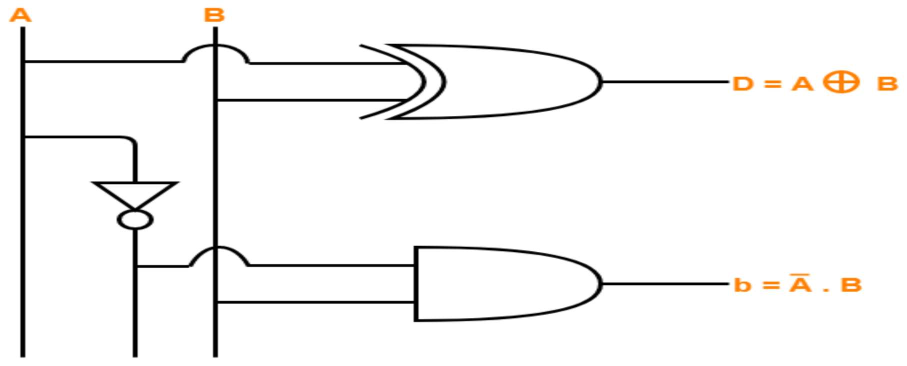


K Maps

Step-04:

Draw the logic diagram.

The implementation of half subtractor using 1 XOR gate, 1 NOT gate and 1 AND gate is as shown below-



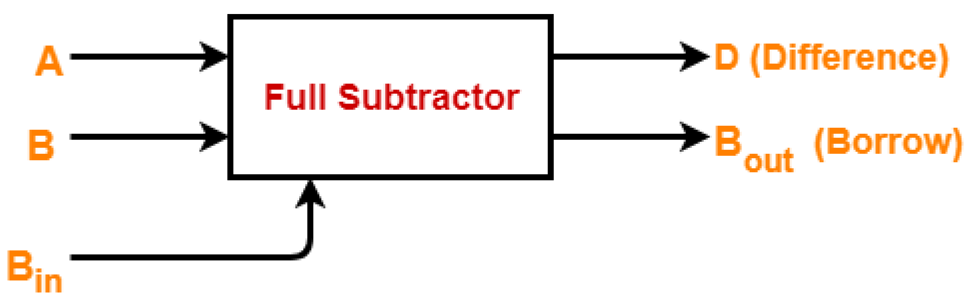
Half Subtractor Logic Diagram

Limitation of Half Subtractor-

- Half subtractors do not take into account "Borrow-in" from the previous circuit.
- This is a major drawback of half subtractors.
- This is because real time scenarios involve subtracting the multiple number of bits which can not be accomplished using half subtractors.

Full Subtractor-

- Full Subtractor is a combinational logic circuit.
- It is used for the purpose of subtracting two single bit numbers.
- It also takes into consideration borrow of the lower significant stage.
- Thus, full subtractor has the ability to perform the subtraction of three bits.
- Full subtractor contains 3 inputs and 2 outputs (Difference and Borrow) as shown-



Designing a Full Subtractor-

Full subtractor is designed in the following steps-

Step-01:

Identify the input and output variables-

- Input variables = A, B, B_{in} (either 0 or 1)
- Output variables = D, B_{out} where D = Difference and B_{out} = Borrow

Step-02:

Draw the truth table-

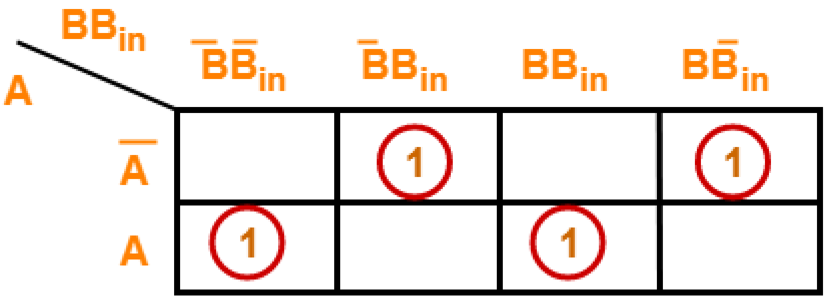
Inputs			**Outputs**	
A	**B**	C	**D (Difference)**	**B_{out} (Borrow)**
0	0	0	0	0
0A’	B’0	C1	1	1
0A’	B1	C’0	1	1
0	1	1	0	1
1A	0B’	C’0	1	0
1	0	1	0	0
1	1	0	0	0
1A	B1	C1	1	1

D=A’B’C+A’BC’+AB’C’+ABC
B=A’B’C+A’BC’+A’BC+ABC

Step-03:

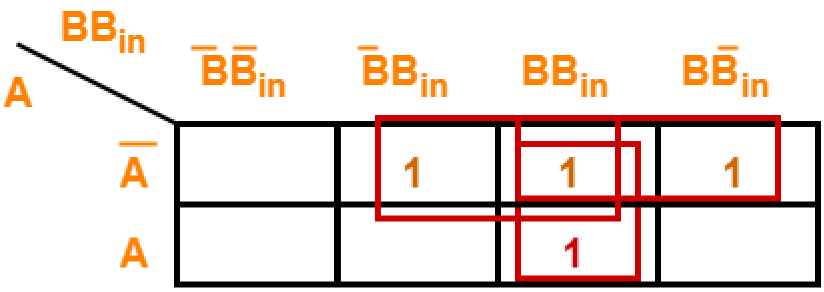
Draw K-maps using the above truth table and determine the simplified Boolean expressions-

For D:



$$\boxed{D = A \oplus B \oplus B_{in}}$$

For B_{in}:



$$\boxed{B_{out} = \bar{A}\,B + (\,\bar{A} + B\,)\,B_{in}}$$

D=A’B’C+A’BC’+AB’C’+ABC

Logical expression for difference –

D = A’B’Bin + A’BBin’ + AB’Bin’ + ABBin

= A’B’Bin+ABBin+A’BBin’ + AB’Bin’

= Bin(A’B’ + AB) + Bin’(AB’ + A’B)

= Bin(A XOR B)’ + Bin’(A XOR B)

Let,

(A XOR B)=X

= BinX’ + Bin’X

= Bin XOR X

= Bin XOR(A XOR B)[(A XOR B)=X]

= Bin (A XOR B)’ + Bin’(A XOR B)

= Bin XOR (A XOR B)

= (A XOR B) XOR Bin

Logical expression for borrow –

B=A’B’C+A’BC’+A’BC+ABC

Bout = A’B’Bin + A’BBin’ + A’BBin + ABBin

= A’B’Bin +A’BBin’ + A’BBin + A’BBin’ + A’BBin + ABBin

=A’B’Bin + A’BBin + A’BBin’+ A’BBin + A’BBin + ABBin

= A’Bin(B’ + B) + A’B(Bin + Bin’) + BBin(A + A’)

= A’Bin + A’B + BBin

OR

Bout = A’B’Bin + A’BBin’ + A’BBin + ABBin

= Bin(AB + A’B’) + A’B(Bin + Bin’)

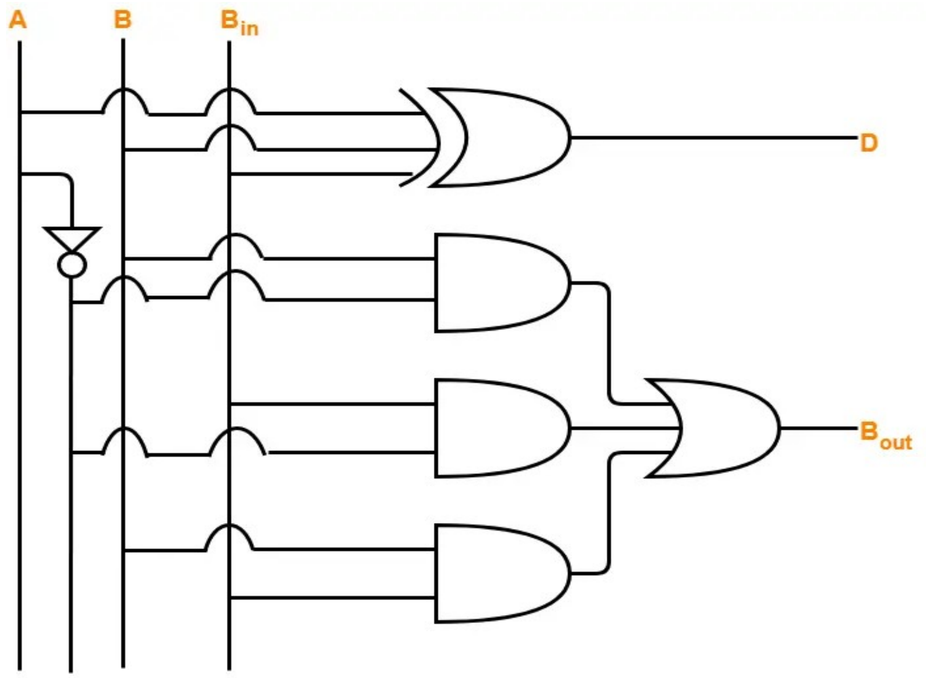
= Bin(A XNOR B) + A’B

= Bin (A XOR B)’ + A’B

Step-04:

Draw the logic diagram.

The implementation of full adder using 1 XOR gate, 3 AND gates, 1 NOT gate and 1 OR gate is as shown below-



Full Subtractor Logic Diagram

ABC

Encoders and Decoders in Digital Logic

Binary code of N digits can be used to store 2^N distinct elements of coded information. This is what encoders and decoders are used for. **Encoders** convert 2^N lines of input into a code of N bits and **Decoders** decode the N bits into 2^N lines.

1. Encoders –
An encoder is a combinational circuit that converts binary information in the form of a 2N input lines into N output lines, which represent N bit code for the input. For simple encoders, it is assumed that only one input line is active at a time.

As an example, let’s consider **Octal to Binary** encoder. As shown in the following figure, an octal-to-binary encoder takes 8 input lines and generates 3 output lines.

Truth Table –

D7	D6	D5	D4	D3	D2	D1	D0	X	Y	Z
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

As seen from the truth table, the output is 000 when D0 is active; 001 when D1 is active; 010 when D2 is active and so on.

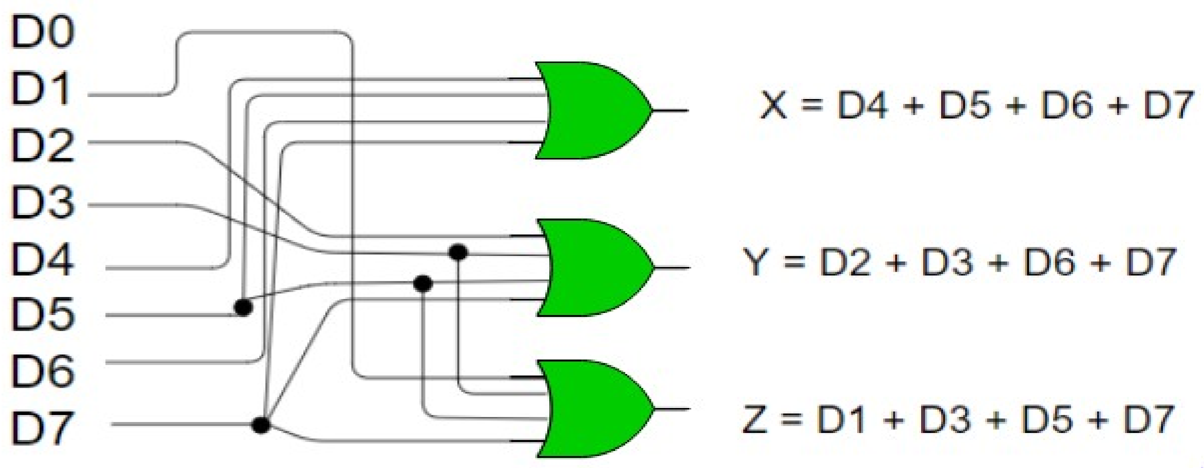
Implementation –
From the truth table, the output line Z is active when the input octal digit is 1, 3, 5 or 7. Similarly, Y is 1 when input octal digit is 2, 3, 6 or 7 and X is 1 for input octal digits 4, 5, 6 or 7. Hence, the Boolean functions would be:

X = D4 + D5 + D6 + D7

Y = D2 +D3 + D6 + D7

Z = D1 + D3 + D5 + D7

Hence, the encoder can be realised with OR gates as follows:



One limitation of this encoder is that only one input can be active at any given time. If more than one inputs are active, then the output is undefined. For example, if D6 and D3 are both active, then, our output would be 111 which is the output for D7. To overcome this, we use Priority Encoders.

2. Decoders –

A decoder does the opposite job of an encoder. It is a combinational circuit that converts n lines of input into 2^n lines of output.

Let's take an example of 3-to-8 line decoder.

Truth Table –

X	Y	Z	D0	D1	D2	D3	D4	D5	D6	D7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Implementation –
D0 is high when X = 0, Y = 0 and Z = 0. Hence,

D0 = X’ Y’ Z’

Similarly,

D1 = X’ Y’ Z

D2 = X’ Y Z’

D3 = X’ Y Z

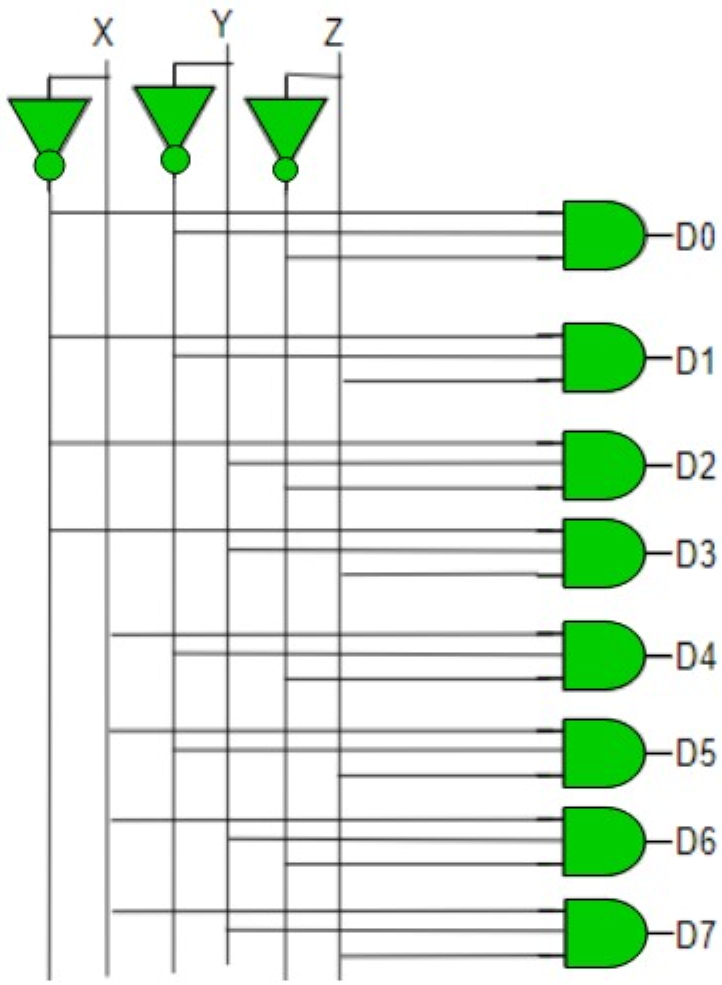
D4 = X Y’ Z’

D5 = X Y’ Z

D6 = X Y Z’

D7 = X Y Z

Hence,


X
Y
Z
D0
D1
D2
D3
D4
D5
D6
D7

The comparison between Encoder and Decoder. We are comparing both terms on the basis of some characteristics to make the comparison clear and more understandable.

S.no.	On the basis of	Encoder	Decoder
1.	**Basic**	It is a combinational circuit that basically converts the information signal to the coded digital bitstream.	It is also a combinational circuit that converts the coded bits to the original information. It performs the reverse operation of the encoder.
2.	**Input lines**	There are $\mathbf{2^n}$ input lines in the encoder.	There are **n** input lines in the decoder.
3.	**Output lines**	There are **n** output lines in the encoder.	There are $\mathbf{2^n}$ output lines in the decoder.
4.	**Operation**	The operation of the encoder is fairly simple.	Unlike encoder, the operation of decoder is complex.
5.	**Basic logic element**	The basic logic element that is used in encoders is **OR gate.**	In decoder, the basic logic element is **AND gate** along with the **NOT gate.**
6.	**Installation**	It is installed at the transmitting end.	It is installed at the receiving side.
7.	**Application**	Encoders are commonly used in Videos, E-mail, etc.	Decoders are commonly used in memory chips, microprocessors, etc.

Both encoders and decoders are the compliment of each other. One is used for encoding the original information, and another is used to decode the coded data bits to get the exact message signal.